Julia Sisk-Reynes

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Supermassive black holes ◊ galaxy mergers ◊ dual & binary active galactic nuclei ◊ accretion ◊ physics beyond the Standard Model ◊ strong gravitational lensing ◊ current and future instrumentation.

EMPLOYMENT AND RESEARCH

Sep 2026 – Present **Postdoctoral Research Associate**
Physics Department, University of Maryland, Baltimore County, Baltimore, MD
Supervisor: Prof. Adi Foord.

Sep 2024 – Aug 2026 **Postdoctoral Research Fellow**
Center for Astrophysics | Harvard & Smithsonian (CfA), Cambridge, MA
"Using gravitational lenses as high-resolution X-ray telescopes" (NASA ADAP).
Supervisor: Dr. Daniel Schwartz.

Jun 2025 – Present **Co-chair of the *NewAthena* Science Working Group 6 (SWG6) on Cosmology & Fundamental Physics**
Chairs: Drs. Esra Bulbul (MPE, Germany) and Florian Pacaud (Bonn, Germany).

EDUCATION

Oct 2020 – Jul 2024 **Ph.D. in Astronomy**
Institute of Astronomy (IoA), University of Cambridge, UK
Thesis: "Unveiling fundamental physics with high-resolution X-ray spectroscopy of active galactic nuclei".
Supervisors: Prof. Christopher Reynolds (University of Maryland, College Park, MD) and Dr. James Matthews (University of Oxford, UK).

Sep 2016 – Jun 2020 **MPhys (BSc + Master's) in Physics** (First Class Honours)
The University of Manchester, UK
Master's thesis: "Building models for the thermal Sunyaev-Zel'dovich effect from analytic and numerical cluster pressure profiles".
Supervisor: Prof. Scott Kay (University of Manchester, UK).

OBSERVING PROPOSALS AND GRANTS

Aug 2026 **PI,** *Chandra* General Observer Program, Cycle 28 – 160 ks **($98,000)**.
PI, *Chandra* General Observer Program, Cycle 28 – 100 ks **($87,000)**.
Targets: GraL J065904.1+16290 and DES J2321-463 lensed quasars.

Aug 2026 **Co-I,** *NASA Astrophysics Data Analysis Programme (NASA ADAP).*
"Probing cosmic growth of supermassive black holes with spin measurements via X-ray reflection spectroscopy". PI: Dr Javier A Garcia (NASA GSFC, MD).

Jul 2024 **Co-I**, Chandra Legacy Program, Cycle 26 – 3 Ms.
Target: Perseus Cluster. PI: Prof. Andrew Fabian (Cambridge, UK).

Jul 2024 **Co-I**, *Chandra* Legacy Program, Cycle 26 – 120 ks.
Target: Zw2089 cluster. PI: Dr Helen Russell (University of Nottingham, UK).

HONORS

Summer 2023	**Flatiron Institute Summer School on Fluid Dynamics** Selected as 1 of 20 from 500+ applicants for the Flatiron Institute's Summer School on Fluid Dynamics. *CCA, Flatiron Institute, New York, NY* (Simons Foundation).
Jan 2022	**Paul Murdin Prize for the best published paper by a current student** From IoA, Cambridge, UK, for Sisk-Reynes et al., 2022, MNRAS, 510, 1
Oct 2020 – Mar 2024	**Full Studentship from the UK Science and Technologies Facilities Council** To pursue a doctorate at the Institute of Astronomy, University of Cambridge, UK.
Jun 2020	**Stellify Award from The University of Manchester** Honorary recognition for extracurricular contributions made whilst an undergraduate student from 2016 – 2020.

PRESENTATIONS

May 2022 – Present: 46 presentations at international conferences, workshops, telescope collaborations, and group meetings. A full list can be accessed here.

- Type: Invited (23), Contributed (23).
- Per year: 2027 (2), 2026 (10), 2025 (11), 2024 (5), 2023 (9), 2022 (9).

Selected recent invited presentations:

Apr 2027	Special Session Talk on black hole spins, APS Global Summit, *Atlanta, GA*
Aug 2026	Representing *NewAthena* SWG6, E1.18, 46th COSPAR Assembly, *Florence, Italy*
Apr 2026	Colloquium, UMBRELA Dialogues, CfA, *Cambridge, MA*
Dec 2025	Seminar, Jeremy Lin Group Meeting, *Hong Kong University*
Oct 2025	Talk on behalf of *NewAthena* SWG6, HEAD/AAS 22 Meeting, *St Louis, MO*
Oct 2025	Colloquium, Universidad Nacional Autónoma de México, *Mexico*
Jan 2025	Colloquium, AGN group, NASA GSFC, *Greenbelt, MD*
Jun 2024	Seminar, Axion Journal Club, Oskar Klein Center, *Stockholm University, Sweden*

SELECTED ACADEMIC SERVICE

Telescope Allocation Time

2025 – 2026	*Chandra* Guest Observer program reviewer, Cycles 27 & 28 (4 days, virtual)
Aug 2025	*XRISM* Guest Observer program reviewer, Cycle 2 (3 days, presential)
Nov 2024	*NICER* Guest Observer program reviewer, Cycle 7 (5 days, virtual)

Journal paper reviewer

Sep 2025 –	Physical Review D, Physical Review Letters, MNRAS

Invited speaker at discussion panels in conferences and meetings

Jun 2024	Dark matter in Astrophysical Laboratories workshop Kavli Institute for Cosmology (KICC), Cambridge, UK
Mar 2024	International Women's Day Discussion Panel for undergraduate students Women in Science Society, University of Cambridge, UK

Conference organization

Jul 2024 — Local Organizing Committee, IoA 50-year anniversary, IoA, Cambridge, UK

SKILLS

Programming	Bash, Python, tcl, Unix/Linux
Astronomy software	(Py)XSPEC, Sherpa, CIAO, ds9, SAOTrace
Numerical tools	ALPRO (axion code, advanced level), AREPO (basic level)
Languages	English, Catalan, Spanish

PRESS COVERAGE

May 2023 — *Athena* Press Release on Sisk-Reynes et al., 2023, ApJ, 930, 1

Jun 2022 — *Chandra* Press Release on Sisk-Reynes et al., 2022, MNRAS, 514, 2
Accompanying Blog Post.
See The Independent, Phys.org, NASA, Space.org, University of Cambridge

STUDENT SUPERVISION AND MENTORING

An asterisk (*) denotes advisee-led and/or advisee-contributed publication.

Dec 2025 – Present — **Jianghao Huyan***. Ph.D. student, University of South Carolina, SC. **Mentor**.
Now: Postdoctoral Research Fellow, Nanjing University, Jiangsu, China.
Project: Optical and IFU observations of strongly lensed quasars.
Publications: 1, paper submitted to ApJL as co-leading author (arXiv: 2609.31826).

Sep 2025 – Present — **Henry Hiscocks***. Exchange master's student, CfA, MA. **Main supervisor.**
Now: Ph.D. candidate, The University of Manchester, UK.
Project: Probing axions constraints with cluster magnetic fields via an MHD simulation of a cluster merger.
Publications: 2 (1 first-author, in prep.; 1 second-author, submitted for the NewAthena Special Issue).

Sep 2025 – Present — **Charlotte Eades***. Exchange master's student at CfA, MA. **Co-supervisor**.
Now: Ph.D. candidate, The University of Oxford, UK.
Project: X-ray-to-optical milliarcsecond astrometry of the Einstein zigzag lens.
Publications: 1 first-author paper (currently in review in ApJ).

Nov 2024 – Jul 2026 — **Alexander Ostridge***. Exchange master's student at CfA, MA. **Supervisor.**
Now: Analyst at Capteiro, UK.
Project: Machine learning application for strong lensing models of quadruply lensed quasars.
Publications: 1 first-author paper (Ostridge et al., 2026, MNRAS, 551, 2).

Jun – Aug 2022 — **Esin Gulhabar**. Summer research student at IoA, Cambridge, UK. **Advisor.**
Now: Ph.D. student at FAU, ECAP, Dr.-Remeis-Sternwarte, Germany.
Project: X-ray observations of the broad-line radio-loud galaxy 4C +74.26.

TEACHING AND STUDENT SUPPORT

Oct 2022 – Jun 2023	Mathematics I Supervisor. Emmanuel College, University of Cambridge.
Jan – May 2021	Astrophysical Fluid Dynamics. Trinity College, University of Cambridge.
Sep 2017 – May 2019	Peer Assisted Study Sessions (PASS) Leader, University of Manchester.

SELECTED OUTREACH ACTIVITIES

Jul 2025	Speaker, Spanish Public Observatory (100 attendees), CfA, Cambridge, MA
Feb 2025	Speaker on visit to a bilingual school (100 attendees), Chelsea, MA
Mar 2024	Speaker, Lindemann Trust Science Day (120 attendees), Cambridge, UK
Mar 2023	Speaker, Lindemann Trust Science Day (100 attendees), Cambridge, UK

OTHER RELEVANT TRAINING

Sep 2025	Spins Workshop, Wake Forest University (2 days), Winston-Salem, NC
Feb 2025	3rd *XRISM* Community Workshop (3 days), College Park, MD
Jul – Aug 2019	Summer research on direct dark matter searches, Manchester, UK

SELECTED INSTITUTIONAL SERVICE

At the Center for Astrophysics | Harvard & Smithsonian, Cambridge, MA

Apr 2025 – Jun 2026	Lead organizer, monthly collaborative discussions, cosmology journal club
Nov 2024 – Jun 2026	Organizer, weekly High Energy Astrophysics Division seminar series

At the Institute of Astronomy (IoA), University of Cambridge, UK

Oct 2021 – Apr 2024	Lead organizer, IoA International Women's Day celebration Responsibilities: keynote speakers, discussion panel (attendees: 100/year).
Oct 2021 – Apr 2024	Lead organizer, Newnham College MCR Seminar Speaker Series Organized a 4-lecture series on the legacies of slavery (attendees: 70).
Jan 2021 – Oct 2023	Representative at events for the general public and prospective applicants
Oct 2022 – Jun 2023	Representative, Computer Users Committee
Dec 2020 – Mar 2024	Representative for 2020 entry Ph.D. students

At the University of Manchester, UK

Oct 2019 – Jun 2020	Class representative for final-year Physics students
Mar 2017 – Feb 2020	Regular volunteer at events organized by the Physics Outreach Society
Mar 2017 – Jun 2020	Lead committee member, undergraduate Nuclear & Particle Physics Society Organized monthly seminars and a 4-day trip to CERN (50 attendees).

COLLABORATIONS AND SCIENTIFIC MEMBERSHIP

May 2025 –	Co-chair, *NewAthena* SWG6 - Cosmology and Fundamental Physics
May 2025 –	Member, *NewAthena* SWG2 - Galaxies and Supermassive Black Holes
Dec 2024 –	Member, *AXIS* Working Group on Active Galactic Nuclei
May 2023 – Oct 2024	Member, *STROBE-X* Working Group on Cosmology
2024 –	American Astronomical Society (AAS)
2024 –	AAS High Energy Astrophysics Division (HEAD)
2024 –	International Astronomical Union (IAU) Junior Membership
2022 – 2024	Royal Astronomical Society